# Yuankun Huang

Software Engineer, Game Systems & Simulation

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Unity/C# client engineer who builds the systems other engineers ship on. Five years in live-service mobile games: core client engineer on Mythic Heroes from early development to 10M+ downloads, now leading the live-event framework on Top Museum. Moving toward real-time simulation through an offline physics bake at work and deterministic replay in C++20.

## CORE SKILLS

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|---|---|
| **Languages** | C# (production since 2020), C++20 (personal projects) |
| **Unity & Client** | Unity (mobile, live service), Timeline/Animator, Addressables, ILRuntime hot-update, Play Mode tests, Unity Profiler, Frame Debugger, editor tooling |
| **Simulation** | Collision detection, offline batch simulation, deterministic replay, multi-agent simulation, spatial partitioning |
| **Networking** | Google Protobuf, SignalR, MessagePack, .NET 8 |
| **Build & Workflow** | CMake, GitHub Actions CI, Cursor (team agentic workflow) |

## PROFESSIONAL EXPERIENCE

**Unity Developer** May 2025 – Present
*IGG Canada, client development on two live-service mobile games* *Vancouver, BC*

**Top Museum, mobile strategy game (test release on Google Play)** Feb 2026 – Present

- Lead the live-event client framework (C#/Unity) on which six other engineers have shipped 30+ event types; new events plug in through a content factory and registries, with no changes to hub, calendar or red-dot code.
- Catalogued ten recurring anti-patterns in the event module and encoded a layering standard in the team's Cursor agent workflow, so AI-generated features stay within the architecture; refactored guild duel as the reference.
- Built 13 event types end to end, including guild duel, demolition, tribe and wolf, and extracted the participation and warm-up base that four guild events now share.
- Designing a pachinko-style event around offline batch simulation: a 2D ball simulator with swept collision detection runs 5M seeded launches and keeps only paths that meet design rules, so designers set odds from measured outcomes.
- Packed 7,693 accepted paths into a 1.1 MB binary track file; the server draws each outcome and the client replays a matching track, so every device shows the same path and runs no physics.
- Wrote Play Mode regression suites for 12 event types (78 cases) on the team's test framework, plus a nightly runner; they caught rewards being dropped after a design change put two in the same second.
- Automated build-size stripping for hero showcase scenes (Timeline/Animator): it runs on every regeneration, compares full and stripped scenes frame by frame and rolls back on any visible difference.
- Built a UI-layer prefab validator (editor tooling) that every UI generator saves through, and mentor new client engineers.

**Mythic Heroes, live-service mobile RPG (10M+ downloads)** Jun 2025 – Dec 2025

- Owned the pet progression line (summon, training, refinement, bonds) and built the flip-card event and event calendar, shipping 25 new UI windows through monthly ILRuntime hot-updates.
- Co-developed the blind-box event, added Protobuf requests for the pet and flip-card flows, and fixed live defects.

*Career Break: Bereavement & Family Care* *Feb 2025 – May 2025*

**Software Engineer (Contract)** Jan 2024 – Jan 2025
*The HIVE, UBC Faculty of Medicine, gamified stroke rehabilitation* *Vancouver, BC*

- Sole engineer: rebuilt TrAIT, a stroke-rehabilitation game with Ultraleap hand tracking, from the ground up, replacing an early prototype; it succeeds the Kinect version validated in *Neurorehabilitation and Neural Repair*.
- Made the game playable for patients with limited arm reach: multi-stage calibration maps each patient's reach to the full screen, and selection works by dwell rather than gestures.
- Built the training rules into the game: streak-based progression across 10 levels and a daily session cap rebuilt from on-device logs, so an app restart cannot bypass it.
- Gave researchers per-frame movement data as CSV logs, zipped and emailed to the lab over SMTP.
- Built a ScriptableObject config pipeline with editor code generation, so the team's designer tuned all 10 levels without code changes; worked with a product owner and that designer.

**Unity Developer** Sep 2020 – Sep 2024

*IGG Canada, Mythic Heroes (live-service mobile RPG, 10M+ downloads)* *Vancouver, BC*

- Core client engineer from early development through global launch and live ops.
- Owned the Chat client end to end: channels, direct messages, emoji, moderation and server-pushed presence over the game's Google Protobuf connection.
- Co-owned the Guilds client, including client networking and multiplayer state sync over Protobuf.
- Co-developed the shared UI framework behind the game's screens and authored 150+ of its windows.
- Kept long lists and leaderboards smooth on low-end Android: profiled with the Unity Profiler and Frame Debugger, then added off-screen list culling and leaderboard caching and moved sprite atlases to Addressables.
- Shipped monthly live content through the ILRuntime hot-update pipeline.

## SELECTED PROJECTS

**Rhythm Fruit Shop [GitHub]** *C++20 / SFML / CMake / GitHub Actions*

- C++20 rhythm-game core with an audio clock derived from the PCM playback cursor; GitHub Actions CI requires that deterministic replay reproduces the recorded result exactly and a perfect run matches a reference scoring implementation.
- A custom operator-new guard asserts that the headless session core's input and update path never allocates; CMake dependency guards fail the build if a core layer links SFML or miniaudio.

**Kodama [GitHub]** *.NET 8 / SignalR / MessagePack / Unity*

- Server-authoritative multi-agent simulation in .NET 8: 10K agents on a 10 Hz tick in SoA component storage, with a behaviour pass measured at zero allocations.
- Uniform-grid spatial partitioning for nearest-resource queries, checked against a linear scan on 60K randomized cases; state streams over SignalR and MessagePack to a GPU-instanced Unity client.

## EDUCATION & CERTIFICATION

**M.S. Computer Science** (in progress, part-time), Georgia Institute of Technology 2026 – Present

**Master of Digital Media**, Centre for Digital Media, Vancouver 2021

**B.Eng. Telecommunication Engineering**, Queen Mary University of London (joint programme with BUPT) 2019

Unity Certified Expert: Programmer